

Literature-Implied Answer: A Limit on Relaxing Weak-Star Compactness

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Status

Status: literature-implied negative answer to the broad relaxation of Theorem `boundary` in arXiv:1302.0160 obtained by deleting the w^* - σ -compactness hypothesis.

Source Question

In arXiv:1302.0160, *Polyhedrality in Pieces*, Fonf, Pallares, Smith, and Troyanski prove Theorem `boundary`: if $E \subset X^*$ is σ - w^* -LRC and w^* - σ -compact and contains a boundary of X , then every $\varepsilon > 0$ admits an ε -approximating equivalent polyhedral norm. Immediately after the theorem they explain that the extra w^* - σ -compactness assumption is used to control the locations of limit points, and that they do not know to what extent it can be relaxed.

Supporting Literature

In arXiv:1804.02899, *Topology, isomorphic smoothness and polyhedrality in Banach spaces*, Smith recalls the standard necessary condition that if a Banach space admits a polyhedral norm, then it is Asplund and c_0 -saturated; see Theorem `thm_nec_conditions`. Smith also records Bible's example

$$Z = \ell_1 \oplus \ell_1(S_{\ell_\infty}),$$

which is neither Asplund nor c_0 -saturated and hence is not isomorphically polyhedral, but nevertheless admits an equivalent norm with a relatively w^* -discrete boundary. The same paper states in Example `weakstarlrcex` that any w^* -relatively discrete subset of a dual Banach space is w^* -LRC.

Identification

Bible's boundary for the renormed space Z is relatively w^* -discrete, so it is w^* -LRC and therefore σ - w^* -LRC using just one piece. It is also a boundary for the chosen equivalent norm on Z .

If Theorem `boundary` from arXiv:1302.0160 remained true after simply removing w^* - σ -compactness, then applying that hypothetical theorem to this renormed Z would produce an equivalent polyhedral norm on Z . This contradicts Smith's necessary conditions, since Z is not Asplund and is not c_0 -saturated. Thus the w^* - σ -compactness hypothesis cannot be dropped outright.

Scope

This is not a new proof of an original counterexample; it is an agent-identified implication from Smith's published discussion of Bible's example. The packet only answers the strongest naive relaxation

$$\sigma\text{-}w^*\text{-LRC boundary} \not\Rightarrow \text{isomorphically polyhedral}.$$

It does not settle finer intermediate relaxations of the source theorem that still impose some replacement for limit-point control, such as countable decompositions whose w^* -closures remain inside the boundary set.

Search Evidence

The cheap run indexes contained one previous attempt for arXiv:1302.0160 and one related literature-implied packet for arXiv:2502.17001. The related packet already identified the same Bible/Smith example as a negative answer to the later weak-star σ -discrete boundary implication. Direct index searches for arXiv:1302.0160, *Polyhedrality in Pieces*, $\sigma\text{-}w^*\text{-LRC}$, and $w^*\text{-}\sigma\text{-compactness}$ showed no prior durable packet for this exact 1302.0160 relaxation target.

References

References

- [1] V. P. Fonf, A. J. Pallares, R. J. Smith, and S. Troyanski, *Polyhedrality in Pieces*, arXiv:1302.0160.
- [2] R. J. Smith, *Topology, isomorphic smoothness and polyhedrality in Banach spaces*, arXiv:1804.02899.
- [3] J. M. F. Castillo and R. Salguero Alarcon, *Polyhedrality for twisted sums with $C(\omega^\alpha)$* , arXiv:2502.17001.